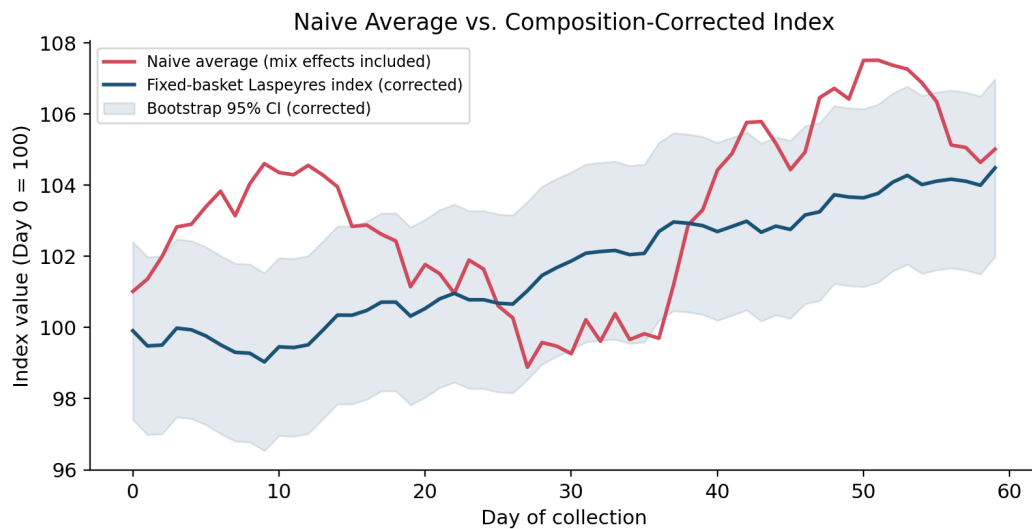


Project APIx

Airfare Price Index for India

A Composition-Corrected Alternative to Naive Fare Averaging



Smart India Hackathon 2026

Problem Statement SIH26056

Ministry of Statistics and Programme Implementation (MoSPI) / DGCA

*“Every naive average confuses two questions:
what did prices do, and what did we happen to look at.
This document is about never confusing those two questions again.”*

Internal Strategy Document — September 26, 2026

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Chapter 1

The Goal

1.1 What We Are Actually Building

We are building **APIx (Airfare Price Index)**: a system that scrapes Indian airline and OTA websites daily, and turns those noisy, dynamic-priced, partially-missing observations into a statistically defensible index of how much flying actually costs in India — separated cleanly from changes in *what happened to be observed*.

The one-sentence goal:

Build the first Indian airfare index that can tell the difference between “*fares actually went up*” and “*we just happened to see more expensive bookings this month.*”

1.2 Why This Problem Statement Exists

The Consumer Price Index (CPI), published monthly by MoSPI/NSO, feeds directly into RBI’s inflation-targeting monetary policy. The ‘Transport and Communication’ sub-group still collects air-fare prices largely through manual, limited-outlet collection – even though over 90% of domestic tickets are now sold online through airlines and OTAs (MakeMyTrip, Yatra, EaseMyTrip, Cleartrip, Ixigo, Goibibo). Manual collection cannot capture fares that swing 200–400% in a single day depending on advance-booking window, day-of-week, and demand surges.

The PS explicitly asks for an automated, high-frequency, scalable system that mirrors what a real Indian traveller actually pays – and asks teams to backtest at least 30 days of results against publicly available DGCA monthly average-fare data.

1.3 Why Most Teams Will Fail This Problem

Out of an estimated 40+ competing teams on this exact PS, the overwhelming majority will submit some variation of:

scrape prices → average them → put on a dashboard → compare loosely to DGCA

This is not wrong, but it is **not an index** in any rigorous economic sense. A simple average of whatever fares you happened to observe conflates two completely different phenomena:

1. **Real price movement** – fares for the same product (same route, same carrier mix, same booking window) genuinely got more or less expensive.
2. **Composition / mix effects** – your average moved because you happened to scrape more long-haul routes this week, or more people in your sample booked last-minute, or a low-cost

carrier added more inventory.

A judge who understands index theory (and SIH panels usually include at least one such person) will ask exactly one devastating question of every naive team: *“How do you know your 8% increase isn’t just because you scraped a different mix of flights this month?”* Naive teams have no answer. We will.

Chapter 2

Our Philosophy

2.1 The Core Insight (Read This Twice)

An index is not an average. An index is a controlled comparison.

The entire discipline of index-number theory (Laspeyres, Paasche, Fisher – all over a century old) exists to solve exactly one problem: how do you compare a “basket” of prices across time when the basket itself keeps changing? CPI, WPI, and every serious price index in the world is built this way. Nobody scrapes prices and simply averages them and calls it inflation. We refuse to do that either.

This is our entire intellectual position, and it is the single thread that must run through every slide, every line of code, and every answer we give a judge. We are not a scraping project with a dashboard bolted on. We are an index-construction project that happens to need a scraper to get raw material.

2.2 Why This Is Defensible, Not Just “Sounds Smart”

We are explicitly *not* claiming to have invented a new algorithm. Laspeyres indexing is 100+ year old economics. Our defensibility does not come from novelty of method – it comes from being the *only team that correctly applied a known, rigorous method to a domain that currently has none.*

What we are NOT claiming:

- We are NOT claiming our index is more “accurate” than DGCA’s in some absolute sense.
- We are NOT claiming AI/ML makes this novel – we use standard statistics correctly, not buzzwords.
- We are NOT claiming our scraper is technologically special.

What we ARE claiming:

- We correct a real, quantifiable, explainable methodological error that a naive average makes.
- We can show, with our own data, exactly how large that error is.
- We can explain, in one sentence, why a fixed basket + lead-time stratification produces a more economically meaningful number.

2.3 The Three Pillars

Pillar 1: Fixed Basket, Not Free Sampling. We decide, in advance, which routes and which booking-window buckets constitute “the basket.” We never let the basket

silently drift based on what the scraper happens to find that day.

Pillar 2: Decomposition Over Assertion. We never just report a number. We always show *how much of the number is real price movement* versus *how much is composition/mix effect*. This decomposition chart is our single most important deliverable.

Pillar 3: Reference, Not Ground Truth. DGCA data is a reference point we compare against and explain divergence from – never a ground truth we claim to match. We will be the only team in the room who does not overclaim DGCA validation.

Chapter 3

Key Terms (The Words You Need to Actually Know)

This is the glossary. If you cannot explain every term in this chapter, in your own words, without looking at this document, you are not ready to present. A judge will use exactly this vocabulary to attack you.

Airfare Price Index (APIx)

Our project's name for the number we produce daily: a single figure (or set of figures per route/segment) tracking how the cost of flying in India is changing over time, expressed relative to a base value of 100 on day one.

Basket

The fixed set of routes (e.g., DEL-BOM, DEL-BLR, BOM-BLR, DEL-CCU, BLR-HYD, MAA-DEL) and booking-window buckets we track every single day, chosen once at the start using DGCA passenger-traffic data, and never silently changed.

Booking Window / Lead-Time Bucket

How many days before departure the fare is observed: T+1 (booking tomorrow), T+7, T+15, T+30, T+45. A T+1 fare and a T+45 fare for the identical flight are economically different products – one is distress/last-minute pricing, the other is advance-planning pricing. Pooling them together is a category error.

Composition Effect / Mix Effect

The part of an observed price change that comes purely from *which* flights you happened to sample – not from any actual price movement. Example: if this week you accidentally scraped more premium long-haul fares than last week, your raw average goes up even if every individual fare stayed exactly the same.

Laspeyres Index

An index formula that computes the cost of a *fixed* basket of goods (here: routes $\times$ booking-windows) using current prices, always weighted by base-period quantities/weights. This is precisely the tool that prevents composition effects from contaminating the number. It is the same family of method used to construct CPI itself.

Normalization

The process of stripping out taxes, User Development Fee (UDF), and convenience charges so that what remains – the base fare – is genuinely comparable across different OTAs and airlines that display these components differently.

Outlier / Missing-Data Handling

Rules for excluding clearly erroneous observations (e.g. business-class fares mixed into an economy

basket, sold-out/cancelled listings) and for statistically handling gaps where a route/window combination simply has no observation on a given day.

Bootstrap Confidence Interval

A resampling technique that lets us report our index not as a single bare number, but as a number with an honest uncertainty band (e.g. “+3.3% $\pm$ 0.8%”). Almost no competing team will report uncertainty at all – this alone is a visible signal of rigor.

Decomposition

Splitting an observed index change into its component causes: how much came from real price inflation, how much from route-mix shift, how much from booking-window mix shift. This is the single chart that makes our submission different from every naive scrape-and-average team.

DGCA Reference Comparison

Comparing our daily/weekly index against DGCA’s publicly released monthly average domestic fares – explicitly framed as *contextual validation*, never as proof of correctness, since DGCA’s own methodology (unstratified, monthly, broad averaging) is structurally different from ours.

Ethical / Rate-Limited Scraping

Collecting data while respecting `robots.txt`, using reasonable request rates, and avoiding aggressive anti-bot circumvention – because the PS explicitly requires this, and because a judge can and will ask about it.

Chapter 4

Working Strategy

4.1 System Architecture

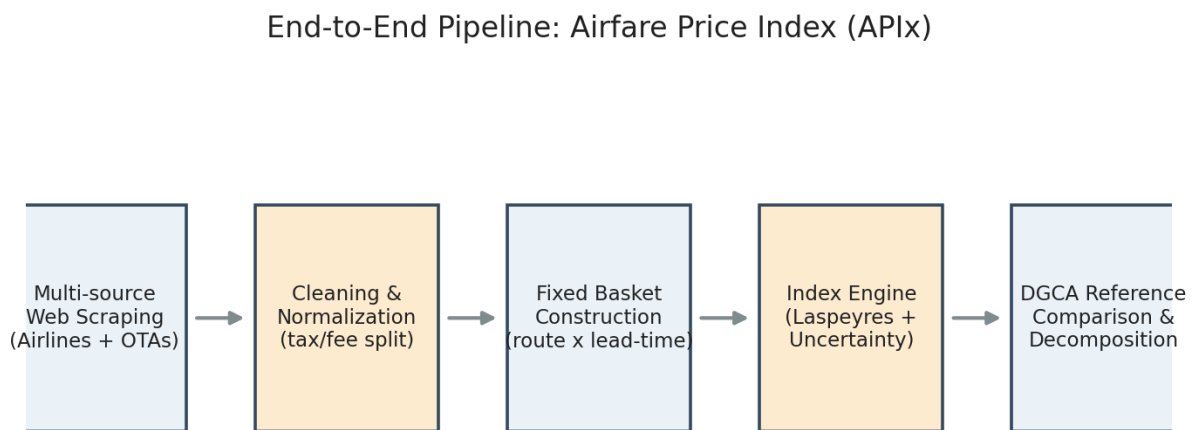


Figure 4.1: End-to-end pipeline. Every box in this diagram must correspond to actual working code by the PPT deadline – not a slide-only concept.

4.2 Stage-by-Stage Breakdown

4.2.1 Stage 1 — Multi-Source Scraping

Collect fares daily from major airlines (IndiGo, Air India, Air India Express, Akasa Air, SpiceJet) and leading OTAs, across the fixed basket of routes and lead-time buckets. Technology choice (Scrapy, Selenium, Playwright) is secondary to *consistency*: the scraper must run every single day without gaps, because gaps are the single biggest threat to backtesting credibility.

4.2.2 Stage 2 — Cleaning & Normalization

Split base fare from taxes/UDF/convenience fees. Drop cancelled/sold-out listings. Flag and handle missing route×window combinations rather than silently ignoring them.

4.2.3 Stage 3 — Fixed Basket Construction

Lock the basket (routes chosen via DGCA traffic-share data, lead-time buckets T+1/T+7/T+15/T+30/T+45) *before* serious data collection begins. This must be decided and documented in Week 1, not adjusted later based on what looks good.

4.2.4 Stage 4 — Index Engine

Compute two numbers side by side, every single day:

- The **naive average** (what every other team will show).
- Our **Laspeyres, composition-corrected index** (fixed weights, held constant).

The gap between these two lines is our entire pitch.

4.2.5 Stage 5 — Uncertainty & Decomposition

Bootstrap a confidence band around the corrected index. Decompose any observed change into: real price movement vs. route-mix effect vs. booking-window-mix effect.

4.2.6 Stage 6 — DGCA Reference Comparison

Overlay DGCA’s monthly average fares against our daily/weekly index, and explicitly explain any divergence via known methodological differences (stratification vs. no stratification) – rather than claiming alignment proves correctness.

4.3 The Two Charts That Carry the Entire Presentation

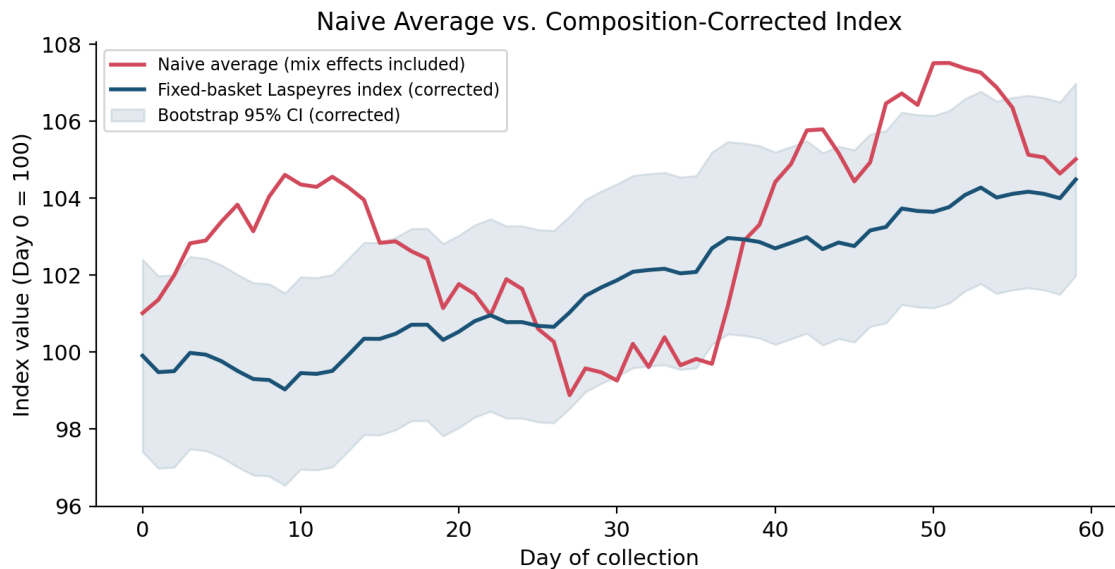


Figure 4.2: Chart A: the naive average (red) vs. our composition-corrected index (blue), with uncertainty band. This single chart demonstrates, visually, that naive averaging overstates or misrepresents true fare movement.

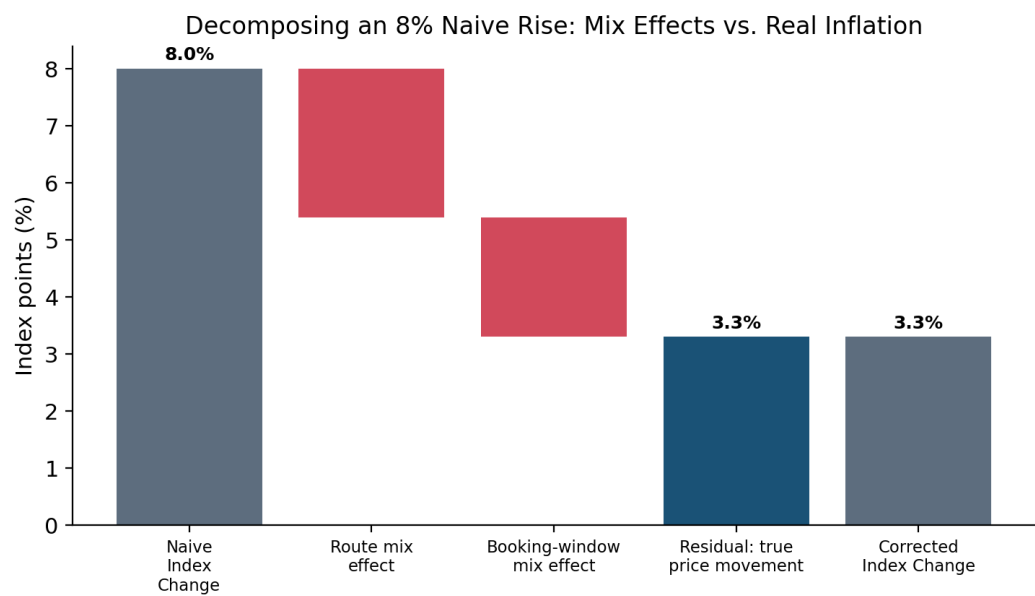


Figure 4.3: Chart B: decomposition of an 8% naive fare increase into route-mix effect, booking-window-mix effect, and residual real price movement. This is the chart that answers “so what did you actually find” in one glance.

4.4 Execution Timeline

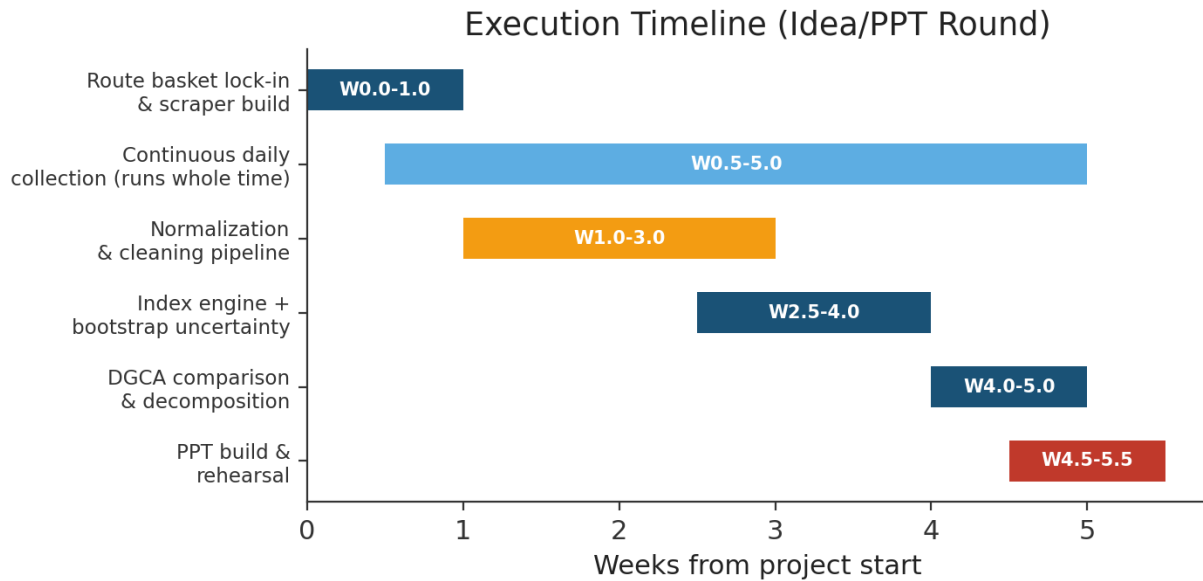


Figure 4.4: Data collection must start in Week 0 – it is the one resource that cannot be compressed later. Everything else can be built in parallel or compressed if needed; days of missed scraping cannot be recovered.

Timeframe	What Must Be Done
Week 0 (immediately)	Lock the route basket and lead-time buckets. Stand up the scraper and start continuous daily collection – this must not stop, even while other stages are being built.
Weeks 1–2	Build and validate the normalization pipeline; confirm base-fare extraction is clean across all sources.
Weeks 2.5–4	Implement both index formulas (naive vs. fixed-basket Laspeyres); produce the divergence chart as an internal proof-of-concept.
Weeks 4–4.5	Add bootstrap uncertainty bands; pull DGCA monthly data and build the reference comparison with explicit divergence explanation.
Final week	Build the PPT entirely around the two core charts (naive vs. corrected; decomposition). Rehearse judge Q&A (Chapter 5) until every answer is instant and unhesitating.

Chapter 5

Fallback Plans

Every part of this plan has a failure mode. Below is what we do when things break — because in a 36-hour-adjacent, scraping-dependent project, something *will* break.

If this fails...	...we fall back to this
An airline/OTA site blocks scraping or changes its layout mid-project	Immediately drop that single source from the basket rather than stalling the whole pipeline; document the gap transparently in the PPT as a known, handled limitation, not a hidden failure.
We don't get 30 full days of clean data before the deadline	Present the index construction methodology as fully built and validated on whatever real days we do have (even 10–15), and explicitly state the backtest will extend automatically once more days accumulate. Never fabricate missing days.
Live scraping demo fails during the actual presentation (network, CAPTCHA, site downtime)	Never rely on a live demo as the primary proof. Pre-record a demo video and prepare a static dataset snapshot as the default presentation mode; live scraping is shown only as an optional bonus, never as the thing the whole talk depends on.
A judge disputes that our DGCA divergence is meaningful rather than just “being wrong”	Have the exact stratification argument ready verbatim (see Chapter 5) – this is the single most likely attack and must never be answered with hesitation.
We can't cover every route DGCA tracks	Be explicit that our basket is a representative subset selected by traffic share, not an attempt to replicate DGCA's full coverage – scope honesty beats overclaiming coverage we don't have.
Team member unavailable / technical blocker close to deadline	The two-chart core (naive vs. corrected; decomposition) is the non-negotiable minimum viable deliverable. Everything else (dashboard polish, API, extra routes) is optional scope that can be cut without damaging the core argument.

The one fallback rule above all others: If forced to cut scope under time pressure, cut *breadth* (fewer routes, fewer OTAs, shorter history) before ever cutting *rigor* (the decomposition, the uncertainty band, the honest DGCA framing). A smaller, rigorous index beats a large, naive one every time a judge asks a real question.

Chapter 6

Judge Defense Script

These are rehearsed, verbatim-ready answers to the most likely hostile questions. Every team member should be able to deliver these without hesitation.

Q: “Isn’t this just a scraper with a dashboard?”

A: “A scraper reports what prices were observed. We report what actually *changed* – holding the observation mix constant. Here’s the gap between the naive number and our corrected number, and here’s exactly why that gap exists.”

Q: “How do you know your composition correction is even right?”

A: “We fix the basket and its weights before collection starts – exactly the same Laspeyres-family methodology MoSPI already uses for other CPI sub-groups. We’re extending the same rigor to a sub-group that currently has none.”

Q: “Your number doesn’t match DGCA’s – so you’re wrong?”

A: “DGCA publishes monthly, unstratified averages. We stratify by booking window because a T+1 fare and a T+45 fare for the same route are not the same economic product. The divergence is the finding, not the error.”

Q: “With 40+ teams on this exact problem, why should we pick you?”

A: “Every other team will show you an average. We are the only team that can show you the difference between an average and an index – and quantify, in points, how much that difference is worth.”

Chapter 7

Closing Statement

“CPI currently treats a last-minute Delhi–Mumbai fare and a booked-45-days-ahead fare as the same data point. We built an index that doesn’t make that mistake – and we can show exactly how much that mistake is currently costing India’s inflation numbers.”

This is the sentence that opens the presentation. Everything in this document exists to make that sentence true, defensible, and impossible for a judge to dismiss as marketing.